MOHAN BABU UNIVERSITY
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SCHOOL OF COMPUTING
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PROJECT ABSTRACT SUBMISSION (2026-27)

Title of the Project	A Digital Twin-Driven Multimodal Framework for Early Fruit Quality Prediction Using Vision-Language Models
Program of Study	B.Tech CSE-AIML
Batch Number	07
Section	B2
Name of the Project Guide	Dr. K. Padmaja
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Abstract:

Early prediction of fruit quality can help farmers make better decisions about crop management, irrigation, nutrient application, and harvesting. However, many existing fruit-quality assessment methods evaluate the fruit only after it has developed, making it difficult to identify its potential quality at an earlier stage. To address this issue, this paper proposes Fruit Twin GPT, a Digital Twin-driven multimodal framework for predicting fruit quality at the individual-plant level. The framework creates a digital representation of the plant by combining flower and leaf images with information such as soil nutrients, weather conditions, irrigation history, plant age, and fruit variety. A Vision Transformer (ViT) is used to analyze the plant images, while a Graph Neural Network (GNN) helps model the relationships between the plant, its growing environment, and management practices. These different sources of information are combined to predict important fruit characteristics, including weight, sweetness (°Brix), firmness, shelf life, and expected harvest date. A Vision-Language Model (VLM) is also included to explain the predictions in a way that is easier for farmers and other users to understand. By focusing on the plant before fruit development rather than assessing the fruit after it has formed, Fruit Twin GPT aims to provide earlier insights into potential fruit quality and support more informed and efficient crop management. The proposed framework explores how Digital Twins and multimodal artificial intelligence can be combined to improve early fruit-quality prediction and contribute to more sustainable agricultural practices.
Keywords— Digital Twin, Fruit Quality Prediction, Multimodal Learning, Precision Agriculture, Vision Transformer, Vision-Language Model.

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